Causal inference: DAGs and difference-in-differences



Data science · University · Lecture 3

In this lecture

- Spot confounding.
- Choose an adjustment set from a DAG.
- Estimate an effect with DiD.

Remember

- Confounders bias raw gaps.
- Never adjust for mediators or colliders.
- DiD rests on parallel trends.

Draw it, then estimate it

1. Warm-up (5 min): Should we close it?
2. Teach (15 min): DAGs and back doors.
3. Draw (15 min): Your own case.
4. Estimate (20 min): DiD by hand.
5. Wrap-up (5 min): One-minute paper.

Let’s discuss

1. What would make you trust the tutoring estimate?
2. Why is a collider dangerous to adjust for?
3. How will you grade assumptions?

Exit ticket

- Draw a three-node DAG and say what you would adjust for.